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Informal Ties and Sales Agent Turnover: Evidence from Randomized Dormitory Assignment

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Informal Ties and Sales Agent Turnover: Evidence from Randomized Dormitory Assignment

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Abstract

Sales agent turnover poses significant challenges for firms that rely on sales agents to maintain customer relationships and generate revenue. This paper studies how informal ties (e.g., shared hometowns) among agents affect turnover. We develop a conceptual framework that distinguishes between the effect of informal ties while tied peers remain present and the incremental effect of losing a tied peer. The framework helps clarify whether informal ties reduce turnover overall and whether they operate primarily through support or competition. Partnering with a leading Chinese medical equipment company, we assemble a unique panel dataset tracking roughly 2,000 agents over 20 months. New hires are randomly assigned to company dormitories, generating exogenous variation in exposure to peers who share background traits such as hometown, dialect, or alma mater. We document three main findings. First, agents assigned to dorm rooms with more same-background roommates stay with the firm longer on average. Second, tied peers reduce early quit risk while they remain with the firm. Third, the departure of a tied peer raises the remaining agent's quit risk. Together, these findings suggest psychological support as the operative mechanism, supported by survey evidence and the absence of productivity spillovers. A back-of-the-envelope calculation suggests that improving social matching among new hires could generate economically meaningful savings in recruiting and training costs.

Keywords:
Informal Ties, Sales Agent Turnover, Dormitory Assignments

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In many customer-facing industries, sales agents (also known as salespeople) are the key marketing channel: they acquire leads, convert purchase intentions, and nurture long-term customer relationships. A persistent challenge, however, is their high turnover rate, which is nearly double the average across the economy.[1] The resulting churn costs firms billions in recruiting and training (Zoltners et al. 2008). Beyond these direct financial burdens, salesperson turnover disrupts the continuity of customer relationships (Bendapudi and Leone 2002), hurts customer loyalty, and ultimately undermines long-term firm performance (Subramony and Holtom 2012). Reducing salesperson turnover is thus not just an HR objective; it is central to firms' marketing effectiveness and growth. One option firms often consider is investing in peer relationships (e.g., coworker support, informal mentoring, and workplace friendships),[2] and the potential role of such relationships has become more salient as growing concern about workplace isolation and weaker social connection. Yet there is little empirical evidence on whether these peer relationships actually reduce turnover.

Our paper studies one important form of such peer relationships—informal ties among sales agents—and asks how these ties shape turnover. We define informal ties as relationships that arise outside the formal organizational hierarchy, often extending beyond work hours, and are facilitated by shared backgrounds (e.g., hometown, dialect, or education). Such ties are an important source of peer influence in sales organizations, where employees face high pressure and frequent rejection. Yet their implications for turnover are theoretically ambiguous. On one hand, informal ties may provide psychological support and facilitate peer learning, thereby reducing quit risk. On the other hand, they may intensify social comparison and competition, thereby increasing quit risk. Moreover, their effect may differ depending on whether tied peers remain present or whether the tie is disrupted by a peer's departure.

We formalize these forces with a simple conceptual framework that distinguishes two

[1] https://hbr.org/2017/07/how-to-predict-turnover-on-your-sales-team

[2] https://www.forbes.com/sites/jasonwalker/2024/11/11/workplace-besties-weighing-the-impact-of-friendships-at-work/

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margins. The first is the *present-tie effect*, which captures how informal ties affect quit risk while the tied peer remains present, through either support or competition. The second is the *peer-departure effect*, which captures any additional short-run effect of a tied peer's exit. A tied peer's departure has two consequences. First, it removes a present tie, thereby reducing the agent's stock of current ties. This component is part of the present-tie margin, because the agent now has one fewer tied peer still present. Second, the departure may generate an incremental shock—for example, making leaving more salient and socially acceptable, consistent with the turnover contagion documented by Felps et al. (2009), or prompting remaining agents to reassess the value of staying (Lee and Mitchell 1994). Distinguishing the present-tie margin from the departure margin matters because the overall effect of informal ties on turnover can mask opposing mechanisms: for example, the same supportive tie may reduce quit risk while the peer is present, yet that peer's departure may raise quit risk through disrupted support. Understanding which margin dominates and whether ties operate through support or competition carries direct implications for how firms should manage informal ties and respond to peer exits.

Answering our research question and mapping the framework to the data are challenging because informal ties are rarely observed in administrative data, and tie formation is often endogenous—for example, agents sort into informal relationships based on latent characteristics that also shape retention. We address these challenges in a collaboration with a large medical equipment company in China, where sales agents cultivate long-term relationships with hospitals and close high-value equipment deals. We compile detailed personnel records for approximately 2,000 sales agents over 20 months, including turnover outcomes and rich demographic information (e.g., hometown, dialect region, and education) to measure informal tie strength. Crucially, we exploit random roommate assignments to mitigate selection biases (Manski 1993). The company provides dormitories for new hires (described in Section Setting and Data), and within each hiring cohort, agents who opt for company housing are randomly assigned to rooms conditional only on gender. This assignment generates

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exogenous variation in initial exposure to informal ties—both the number of same-cohort roommates and whether those roommates share key background characteristics.

This setting allows us to implement three empirical tests that map directly to the conceptual framework. First, we estimate the intent-to-treat effect (ITT) of initial informal-tie exposure induced by random dorm-room assignment on subsequent tenure. This baseline estimate combines the influence of informal ties while peers remain present with any later effects triggered by peer departures. Second, to isolate the present-tie effect, we focus on the early onboarding period, when peer exits are rare. We estimate how initial tie exposure shapes agents' short-run quit risk. Third, to isolate the peer-departure effect, we estimate how a tied peer's departure changes the focal agent's quit risk, after controlling for the agent's remaining stock of present ties.

We begin by estimating the baseline ITT effect of informal-tie exposure. Specifically, we ask whether new hires who are randomly assigned to rooms with roommates sharing stronger informal ties remain with the firm longer. We find that one additional same-cohort roommate increases tenure by roughly 16 percent. Using alternative proxies for informal ties based on shared background—including shared hometown province, shared dialect, and shared college province—we obtain economically meaningful retention effects of similar magnitude. However, although this positive effect may suggest a support effect, the ITT estimate reflects the combined influence of both the present-tie effect and the peer-departure effect over time, and therefore does not by itself identify whether the net retention effect operates through support or competition.

Our second test isolates the present-tie effect. We focus on the early onboarding period and estimate how exposure to informal ties affects short-run quitting using a discrete-time hazard model. Because roommate composition is randomly assigned at entry and peer exits are rare during this period, differences in quit risk across agents with different tie exposure identify the present-tie effect. We find that stronger informal ties reduce early quit risk, suggesting that informal ties operate primarily as a source of support in our setting.

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Our third test isolates the peer-departure effect. We estimate a discrete-time hazard model that relates an agent's quit risk to the departure of a tied roommate in the previous two weeks. The identifying assumption is that, conditional on agent fixed effects and common time shocks, the exact week in which a tied roommate quits is unrelated to unobserved shocks to the focal agent's quit risk over the subsequent two weeks. We find that the departure of a tied roommate increases the focal agent's quit probability by 8.1 percentage points, with larger effects when the departing roommate is a stronger tie (e.g., 12.7 percentage points for shared hometown). This pattern is consistent with a support-based interpretation: even after accounting for the reduction in an agent's stock of present ties, a tied roommate's departure generates an additional increase in quit risk—suggesting the departure carries a psychological disruption beyond the mechanical loss of a supportive peer.

Across all three tests, the evidence points to a clear conclusion: support, rather than competition, is the dominant influence of informal ties in our dormitory setting. Informal ties reduce quit risk while tied peers remain present, and their disruption raises quit risk after a peer departs. Overall, the positive effect on tenure suggests that, over the observed horizon, the *supportive* effect of tied peers while present outweighs the disruption caused by their departure.

Next, we provide direct evidence on the mechanism underlying these retention effects. The preceding findings are consistent with the idea that informal ties increase retention primarily by providing support. Dorm roommates may mitigate the stress of a demanding, unfamiliar job by offering day-to-day companionship and a sense of belonging (Felps et al. 2009). We field a survey of current agents to provide additional evidence on this psychological support channel. Respondents most frequently identify a roommate, rather than a teammate, as their closest friend at work. They also describe workload pressure as the most difficult aspect of the job and report turning to roommates when they need support. These responses align with the view that informal ties improve retention by buffering stress and strengthening social attachment (Methot et al. 2016).

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Psychological support may also reduce turnover indirectly through early performance, since better adjustment could improve initial productivity, which is a key predictor of turnover (Griffeth et al. 2000). If this channel were first-order, agents assigned stronger informal-tie exposure should perform better early in their tenure. Similar to our first test, we examine whether initial informal-tie exposure increases first-month performance. The estimated effects are small and not statistically different from zero. These results suggest that dormitory informal ties reduce turnover primarily through *direct* psychological support, with no evidence of an *indirect* effect through early performance.

Lastly, we explore whether these retention benefits can extend to formal work teams. Since 2022, the firm has assigned new hires to teams by lottery to curb favoritism and to equalize talent across managers.[3] Because dorm and team assignments are independent, roommates are rarely on the same team. Within our conceptual framework, this comparison is useful because the effect of informal ties depends on the relative strength of support and competition. Unlike dormitories, teams operate within a formal hierarchy and are directly linked to performance evaluation, so informal ties in teams may provide less support and generate more competition. We repeat the baseline ITT analysis using team-level informal ties—shared hometown, dialect, or college with teammates or their team leader. We find no evidence that team-level informal ties affect tenure: estimates are centered near zero. These null results are consistent with the idea that the supportive effects of informal ties are weaker in formal teams, where competitive forces may be stronger. Together, these findings highlight a key implication of our framework: whether informal ties improve retention depends on the organizational context in which those ties operate.

Our study makes two contributions. First, we develop a framework that distinguishes between the effect of informal ties while tied peers remain present and the effect of peer departures. By separating these two margins, the framework provides a tractable way to

[3]Our main analysis focuses primarily on dormitories rather than formal teams for two reasons: (1) dormitory settings provide cleaner identification, and (2) dormitory assignment manipulation is more managerially feasible than team reassignment, since the latter could create workplace conflicts and perceptions of favoritism.

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understand when informal ties matter, when they do not, and whether they operate through support or competition. As importantly, the framework provides a practical diagnostic roadmap for firms: by identifying which margin dominates and whether ties operate through support or competition, firms can determine both whether to cultivate informal ties and how to respond when they are disrupted.

Second, to the best of our knowledge, we provide the first causal evidence on the role of informal ties in sales agent turnover. Whereas the existing salesforce peer-effects literature focuses primarily on formal workplace interactions and productivity (Chan et al. 2014*b*; Mas and Moretti 2009; Li et al. 2020), we study informal ties and retention using quasi-experimental variation from a dormitory assignment lottery. We find that stronger assigned informal ties increase tenure on net, while a tied peer's departure raises the focal agent's short-run quit risk. We further show that these effects arise in dormitories—where informal ties plausibly provide support—but not in formal work teams.

Our findings carry direct managerial implications for salesforce retention. A back-of-the-envelope calculation (Section Managerial Implications) indicates that systematically assigning newcomers with shared backgrounds to the same room would reduce recruiting and training costs by roughly 8.7% of annual profits. More broadly, our results suggest that retention depends not only on who firms hire, but on how they structure the post-entry social environment. Crucially, because the appropriate intervention depends on which margin dominates in a given setting, our framework gives firms a structured approach to diagnosing and acting on the social foundations of employee retention.

The remainder of the paper is structured as follows. Section Related Work and Conceptual Framework reviews the relevant literature and develops our conceptual model. Section Setting and Data describes the institutional setting and the data. Section Empirical Strategy presents the empirical strategy and main results. Section Discussion discusses underlying mechanisms and managerial implications. Section Conclusion concludes.

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RELATED WORK AND CONCEPTUAL FRAMEWORK

Related Work

Our paper contributes to the extensive literature on salesforce management. Most research studies how to improve sales agent performance.[4] Empirical evidence on turnover is much scarcer because personnel-level exit data are rarely available. Many prior studies (e.g., Wotruba and Tyagi 1991; Futrell and Parasuraman 1984) examine turnover *intentions* rather than actual departures. Yet turnover can be as important as performance for sales agents: the exit of a salesperson can jeopardize customer relationships (Bendapudi and Leone 2002) and reduce consumers' future purchases (Subramony and Holtom 2012). Among studies that analyze realized turnover, the identified predictors—individual performance (Trevor et al. 1997), job attitudes (Lucas Jr et al. 1987), external opportunities (Blau 1993), and organizational culture (Sheridan 1992)—are drawn largely from correlational analyses. The study most closely related to ours is Sunder et al. (2017), which shows that peer turnover increases a salesperson's likelihood of leaving. We extend this literature by providing causal evidence that informal ties are an economically meaningful driver of realized turnover.

Peer effects have also attracted growing attention in the salesforce literature, although the emphasis has been on formal workplace structures. For example, knowledge sharing among peers raises productivity (Chan et al. 2014*b*), compensation schemes shape the extent of peer interaction (Chan et al. 2014*a*), and team composition affects individual performance (Xu et al. 2024). Recent evidence outside the salesforce context points to a similar departure-based spillover mechanism. Using data from a fitness club, Chen and Mehta (2025) show that user churn can propagate through consumer networks. Our paper brings this spillover logic into the salesforce setting by studying how peer departures propagate through informal ties and by developing a framework that distinguishes between present-tie and peer-departure

[4]For example, the literature examines compensation structures (Jindal et al. 2022; Lal and Staelin 1986), contract design (Salazar et al. 2025), pricing delegation (Mishra and Prasad 2004; Bhardwaj 2001), and team assignment (Skiera and Albers 1998).

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margins.

Methodologically, our identification strategy builds on the "random-roommate" literature, which exploits random dormitory assignment to isolate peer effects. Prior work shows that who you live with shapes academic outcomes (Stinebrickner and Stinebrickner 2006; Zimmerman 2003), social behaviors such as alcohol use (Kremer and Levy 2008; Sacerdote 2001), long-term incomes (Einiö 2023), and attitudes toward peers (Boisjoly et al. 2006). Our dataset provides rich detail on peer identity: for every roommate pair we observe shared hometown, dialect, and college province. This granularity allows us to examine how different dimensions of informal ties shape turnover.

Table 1 summarizes the related literature along five dimensions: methodology, sales-agent sample size, whether the study examines within-organization peer effects, whether it focuses on informal social ties, and whether it measures realized turnover behavior.

Conceptual Framework

We develop a simple conceptual framework that organizes two distinct margins through which informal ties may shape turnover, and two competing mechanisms—support versus competition—that determine the direction of each margin's effect.

Before introducing the formal model, it is useful to illustrate the two margins with a concrete example. Consider two new sales agents who enter the firm at the same time. Agent A is assigned to a dorm room with two strong-tie peers (e.g., from the same hometown), while Agent B is assigned to a room with one strong-tie peer. In the first few weeks, suppose no one leaves. Agent A is then more exposed to strong informal ties because more of her ties remain present. This is the present-tie margin: retention depends on how many strong ties remain present. If support dominates competition, Agent A is more likely to stay; if competition dominates, Agent B is more likely to stay. Now suppose that in a later week, exactly one of Agent A's strong-tie roommates exits the firm, while Agent B's roommate stays. At this point, A and B have the same number of present ties — but Agent A also faces

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Table 1: Related Salesforce Literature

Study	Study Focus	Methodology	Sample	Peer Interactions	Informal Ties	Turnover Intention	Actual Turnover
Traditional Salesforce Topics (Outcome Variable: Performance)							
Lal and Staelin (1986)	Compensation structures	Analytical modeling	✗	✗	✗	✗	✗
Salazar et al. (2025)	Contract design	Fixed effects model	536 salespeople	✗	✗	✗	✗
Mishra and Prasad (2004)	Pricing delegation	Analytical modeling	✗	✗	✗	✗	✗
Skiera and Albers (1998)	Team assignment	Optimization modeling	✗	✗	✗	✗	✗
Kar et al. (2023)	Customer relationship management	Quasi-experiment	115 salespeople	✗	✗	✗	✗
Turnover Research							
Futrell and Parasuraman (1984)	Effects of performance & satisfaction	Survey methods	263 salespeople	✗	✗	✓	✗
Wotruba and Tyagi (1991)	Effects of expectations	Survey methods	175 salespeople	✗	✗	✓	✗
Lucas Jr et al. (1987)	Effects of demographics & attitudes	Survey methods	1000+ salespeople	✗	✗	✗	✓
Sunder et al. (2017)	Effects of self & peer factors	Hazard model	1,996 salespeople	✓	✗	✗	✓
Peer Effects Research							
Chan et al. (2014*a*)	Peer effects in compensation systems	Structural modeling	61 salespeople	✓	✗	✗	✗
Chan et al. (2014*b*)	Peer-based learning	Quasi-experiment	92 salespeople	✓	✗	✗	✗
Pierce et al. (2021)	Peer bargaining in teams	Fixed effects model	571 salespeople	✓	✗	✗	✗
Xu et al. (2024)	Peer complementarity in teams	Structural modeling	484 salespeople	✓	✗	✗	✗
Our Research Contribution							
Our Study	**Informal ties & turnover**	**Large-scale natural experiment**	**1,987 salespeople**	✓	✓	✗	✓

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the temporary disruption of a peer's departure. This is the peer-departure margin: holding fixed the mechanical loss of one present tie, a peer's departure may generate an additional effect on the remaining agent's retention.

If informal ties operate through support, Agent A—with more present ties—is more likely to stay, and her quit risk rises when a tied roommate departs. If instead ties operate through competition, the pattern reverses: Agent A faces stronger rivalry and is more likely to quit, while her tied roommate's departure may actually provide relief, lowering her quit risk.

We now formalize this intuition. We model an agent's continuation decision as depending on two state variables: the stock of tied peers who remain present and recent exposure to a tied peer's departure. Let the net value of staying for agent i in week t be

$$U_{it}^{S} = u_0(t,i) + \rho\, R_{it} - \lambda\, L_{it}, \tag{1}$$

with quitting normalized to a threshold of 0, so that agent i quits in week t if $U_{it}^{S} < 0$. $u_0(t,i)$ collects all non-tie determinants of staying. Let $\mathcal{P}_i$ denote agent i's roommates, and let $z_{ij} = 1$ if agents i and roommate j share a strong tie. We define agent i's total tie strength at entry as $Z_i = \sum_{j \in \mathcal{P}_i} z_{ij}$.

To capture the present-tie margin, we define the *remaining informal tie strength (present tie)* in week t as:

$$R_{it} = \sum_{j \in \mathcal{P}_i} z_{ij} \cdot \mathbf{1}\{j \text{ is still present at } t\} \tag{2}$$

R_{it} denotes the number of agent i's strong-tie roommates who are still present in week t. To capture the incremental disruption associated with losing close peers, we define *loss exposure* in week t as

$$L_{it} = \sum_{j \in \mathcal{P}_i} z_{ij} \cdot \mathbf{1}\{j \text{ exited within the past } K \text{ weeks}\}. \tag{3}$$

L_{it} denotes the number of agent i's strong-tie roommates who have exited within the past

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K weeks. Because a peer's departure may affect the focal agent with a short delay rather than only in the exact week of exit, we let recent departures matter for K weeks.

Interpretation of parameters. The parameter ρ captures the *present-tie effect*: how retention changes when an agent has more strong informal ties currently present. If informal ties operate primarily through support, then $\rho > 0$. Present ties can provide emotional support that buffers stress and raises the perceived cost of exit (Mitchell et al. 2001; Cohen and Wills 1985), and they can also facilitate information sharing and learning (Chan et al. 2014*b*; Cross and Cummings 2004; Reagans and McEvily 2003).[5] By contrast, if informal ties operate primarily through competition, then $\rho < 0$. In that case, stronger ties may intensify peer competition and social comparison, weaken emotional support, and discourage knowledge sharing (To et al. 2020; Kilduff et al. 2010).

The parameter λ captures the *peer-departure effect*: the additional effect of a tied peer's exit on retention, beyond the mechanical change in the agent's stock of present ties (R_{it}) caused by that exit. If informal ties operate primarily through support, then $\lambda > 0$. The departure of a close peer may create a salient disruption that goes beyond simply having one fewer supportive peer present. It can make quitting more cognitively salient through turnover contagion (Felps et al. 2009), increase stress and distrust among remaining workers (Mishra and Spreitzer 1998), or weaken workplace attachment and self-regard (Brockner et al. 1985), thereby raising quit propensity. By contrast, if informal ties operate primarily through competition, then $\lambda < 0$. In that case, a rival's departure may generate additional psychological relief beyond the effect already captured by the induced change in R_{it}. By removing a salient comparison target from the agent's immediate environment, the exit can reduce interpersonal rivalry and perceived pressure (Kilduff et al. 2010), thereby lowering quit propensity.

Our framework yields four possible configurations, summarized in Figure 1. The sign

[5]For example, peer networks can facilitate the exchange of tacit information about customer preferences, selling techniques, and product features (Chan et al. 2014*b*).

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pattern (ρ, λ) helps identify the dominant mechanism: support predicts $(\rho > 0, \lambda > 0)$, while competition predicts $(\rho < 0, \lambda < 0)$. Mixed patterns—$(\rho > 0, \lambda < 0)$ or $(\rho < 0, \lambda > 0)$—would suggest that different mechanisms dominate across the two margins.

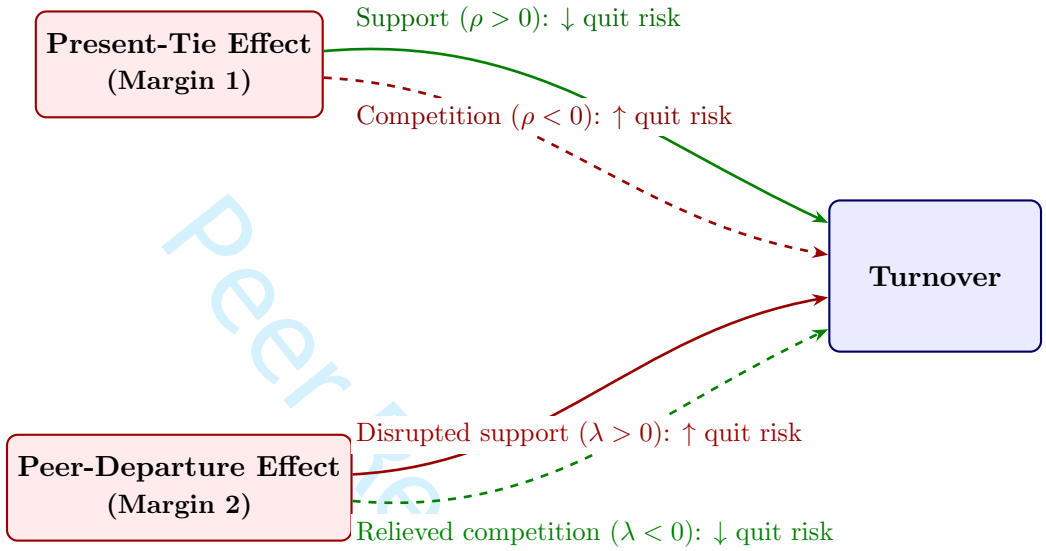

Figure 1: Conceptual Model: Two Margins × Two Mechanisms.

Implications for empirical design. The framework yields three tests that guide our empirical analysis.

Test 1 (Baseline ITT effect on overall tenure). Because stronger initial exposure Z_i implies both greater present-tie strength early on and greater potential exposure to peer departures later, the reduced-form effect of Z_i on overall tenure combines both margins. Taking expectations over the paths $\{R_{it}, L_{it}\}_{t \geq 1}$, the effect of a marginal increase in tie intensity can be approximated as

$$\frac{\partial \mathbb{E}[\text{Tenure}_i]}{\partial Z_i} \propto \rho \cdot \sum_{t \geq 1} \frac{\partial \mathbb{E}[R_{it}]}{\partial Z_i} - \lambda \cdot \sum_{t \geq 1} \frac{\partial \mathbb{E}[L_{it}]}{\partial Z_i}, \tag{4}$$

so the total effect of initial tie exposure on tenure reflects two cumulative forces: (i) the effect of stronger present ties over time, and (ii) the effect of greater exposure to peer-departure shocks. For example, under a support mechanism $(\rho > 0, \lambda > 0)$, stronger initial exposure

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may increase tenure early on through the present-tie margin, but part of this benefit may later be offset if closer peers depart and are also more likely to generate departure shocks. More generally, the sign of the overall ITT does not by itself identify the dominant mechanism, because the reduced-form effect aggregates both margins, and either support or competition effects could dominate. This motivates the more margin-specific tests below.

Test 2 (Present-tie effect). In our setting, peer exits are relatively infrequent during the initial onboarding period, so for $t \leq W$ we have $L_{it} \approx 0$ and $R_{it} \approx Z_i$. Equation (1) then simplifies to

$$U_{it}^S \approx u_0(t,i) + \rho\, Z_i, \qquad t \leq W, \tag{5}$$

so differences in early quit risk across agents with different tie exposure Z_i primarily reflect the present-tie margin and therefore identify the sign of ρ. Under support, $\rho > 0$ implies that stronger ties reduce early quitting. Under competition, $\rho < 0$ implies the opposite.

Test 3 (Departure effect). When a peer j exits, the agent i experiences two changes simultaneously: (i) a mechanical reduction in the present-tie network, captured by ρz_{ij}, and (ii) an additional short-run departure shock, captured by λz_{ij}. The change in the value of staying in the post-exit window is therefore

$$\Delta U_{it}^S\big|_{j \text{ exits}} = -\rho\, z_{ij} - \lambda\, z_{ij}. \tag{6}$$

This motivates estimating how quit risk changes in a short window following the exit of a strong-tie roommate. To isolate the departure-shock parameter λ, we control for the contemporaneous stock of ties that remain present (e.g., R_{it}). Without this control, the estimated post-exit effect would combine the mechanical loss of present ties with the additional shock generated by the exit event itself.[6] Under support, $\lambda > 0$ implies that losing a peer raises

[6] To illustrate, suppose each strong-tie peer contributes one unit to the present-tie network, with $\rho = 2$ and $\lambda = 3$. If an agent initially has two strong-tie roommates, her staying utility before any exit is $U^S = u_0 + 2 \times 2 = u_0 + 4$. If one peer exits, the present-tie network falls from 2 to 1, and the agent is also exposed to a recent departure shock, so $U^S = u_0 + 2 \times 1 - 3 \times 1 = u_0 - 1$. The total decline of 5 units therefore combines a mechanical loss of present ties (-2) and an additional departure shock (-3). Controlling for the remaining stock of ties isolates this latter component.

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quit risk. Under competition, $\lambda < 0$ implies that a peer's exit provides relief and reduces quit risk.

SETTING AND DATA

Institutional Background

We collaborate with a large medical equipment manufacturer and distributor (hereafter "MedTech") in China, specializing in high-value medical devices. Around 70% of MedTech employees are sales agents. MedTech centralizes its nationwide sales operations at its Jiangsu campus, an industrial park comprising large office complexes and dormitories. Similar to university students, sales agents can both live and work within the same campus, facilitating regular interactions among peers.

The company faces significant turnover challenges: it hires roughly 1,200 sales agents each year, yet the average new hire stays only seven weeks. Internal estimates put recruitment and training costs at around 25,000 RMB (around $3500) per agent, so every early exit represents a substantial financial loss.

Figure 2: Sales Agent Process Flow

The journey of new sales agents at MedTech involves five stages: recruitment, random dorm assignment, training, random team assignment, and eventual integration into the sales force (see Figure 2). Due to high turnover, MedTech frequently recruits new sales agents every month. Upon recruitment, around 60% of new hires opt to reside in company-provided dormitories, a common practice among large Chinese firms to streamline onboarding and foster organizational cohesion.

Dormitories typically accommodate four individuals per room (see Figure 3). Rooms

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are randomly assigned by dorm managers on a first-come, first-served basis, subject only to bed availability (illustrated in the left panel of Figure 5). The dorm managers work independently of HR, so room assignments remain independent of individual characteristics (except gender) and are unrelated to future formal team assignments. Roommates often end up on different sales teams.

Figure 3: Dormitory Layout

Note: This figure shows a partial view of a dormitory room at the company campus. Each room contains bunk beds, with the upper bunks typically used for storage rather than sleeping. Rooms generally accommodate up to four residents.

Dorm living can quickly turn strangers into friends. Our small-scale interviews and surveys suggest that many employees consider their roommates among their closest friends within the company (see survey details in Table 8). Agents describe how bonds form through daily routines: sharing meals in the cafeteria, planning weekend hikes, or exploring local malls. These ties offer crucial peer support, especially for newcomers adjusting to an unfamiliar setting. Several respondents note that roommates from the same hometown helped them manage homesickness and navigate early workplace hurdles.

After dorm placement, all newly recruited agents in a given cycle attend a centralized, one-week training program. Throughout the paper, we refer to agents who train together as belonging to the same "cohort". After training, new hires are allocated by HR to one

of approximately 40 sales teams based on team vacancies. Each team typically consists of around 14 active sales agents supervised by one team leader. Since 2022, the team assignment process has also been randomized. Company leadership explicitly adopted this policy to promote fairness and prevent favoritism (e.g., supervisors hand-picking preferred recruits). We verify compliance by checking that predetermined employee characteristics are uncorrelated with team composition, and we find no evidence of systematic sorting (Appendix Section Random Assignment to Teams).

In the main analysis, we focus on dorm-based informal ties rather than team-based ties because the dormitory setting provides cleaner identification: assignments are fully random conditional on gender and independent of team placement. We later use the team setting as a boundary test of the framework, asking whether the dormitory findings extend to formal teams where the balance between support and competition may differ.

Sales agents' main task is to place outbound calls to prospective clients, typically hospitals, convert interested contacts to in-person visits, and ultimately close deals. Because the firm sells high-value medical devices (from basic digital radiography systems to robotic surgical units priced between 100,000 and over 1 million RMB), the conversion rate from first call to final sale is inevitably low. Consequently, during a hire's first few months, performance reviews and bonuses hinge on intermediate metrics (e.g., call volume and call quality). Our performance analysis focuses on these intermediate measures that provide informative statistics on early performance differences across agents.

Data

We use proprietary personnel records provided by the company, capturing detailed information on 1,987 sales agents recruited between January 2023 and August 2024. The dataset includes employee demographics, residence details, and performance metrics. This dataset offers two key advantages for studying turnover. First, the large sample and complete entry-to-exit histories provide sufficient variation in both quitting behavior and roommate com-

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position. Second, detailed demographics, including hometown province, dialect region, and college location, allow us to measure informal ties along multiple meaningful dimensions.

Demographics. The demographic data include each sales agent's gender, age, marital status, and education level. We present basic demographic statistics in Table 2 Panel A. More importantly, we observe geographic information indicating the sales agents' hometown provinces, as well as the provinces where they attended college if applicable. Leveraging their hometown information, we construct a dialect region measure.[7] Agents originating from neighboring provinces typically speak similar dialects, reflecting closer cultural similarity.

Figure 4 maps agents' hometown provinces. Although there is considerable geographic variation, most recruits come from provinces adjacent to Jiangsu, where the firm is headquartered. The same pattern emerges when we group by dialect region or college province (Appendix Figures E1 and E2). This geographic clustering increases the odds that roommates share cultural backgrounds, generating the variation in informal ties that we will exploit.

Dorm residence. We obtain dormitory residence records covering 1,412 (approximately 60%) of the sales agents who opted into the dormitory arrangement. These records include each agent's actual room number, check-in date, and check-out date. Using this information, we can reconstruct the living arrangements within each room (i.e., who lives with whom) at any given time during our study period. On average, dorm residents stay for approximately 57 days, aligning closely with their overall tenure (51 days) at the company.

To better describe dorm residency, we define each observation by a combination of room ID and the first day of each month (e.g., room 301 on July 1st), referred to as a room-month. There are 67 unique rooms, each with a full capacity of four residents, although the actual occupancy rate varies monthly. Across the 1,340 room-month observations (67 rooms $\times$ 20

[7] We classify dialect regions using a standard linguistic system that divides Mandarin and other Chinese language varieties into 11 major dialect groups common in Chinese sociolinguistics. In our sample, Jianghuai Mandarin speakers are most prevalent (51.6%), followed by Zhongyuan Mandarin (16.9%) and Southwestern Mandarin (8.0%).

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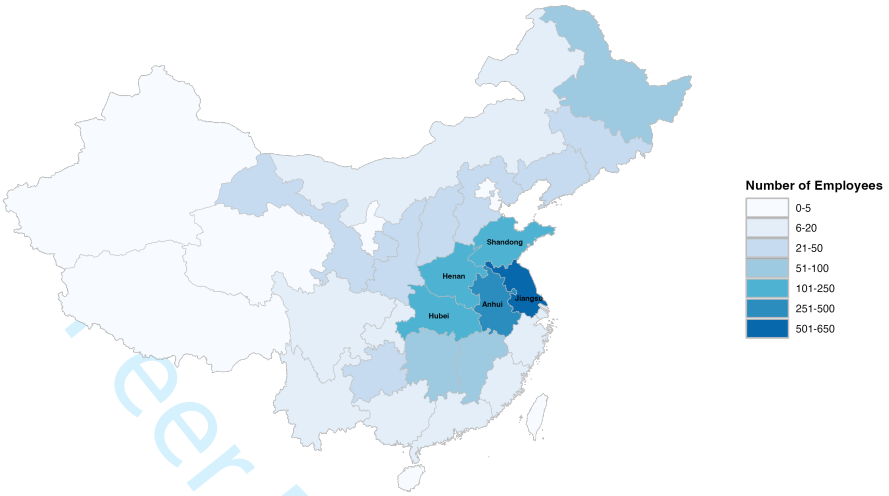

Figure 4: Employee Distribution by Hometown Province

months), we observe substantial variation in occupancy, with many room-months housing fewer than four occupants.

Table 2 (Panel C) reports four measures of informal ties. On average, an agent shares a dorm room with about 1.5 same-cohort peers. We further classify these ties along three dimensions: shared hometown province, dialect region, and college province. Specifically, the average agent has 0.25 same-cohort roommates from the same hometown province, 0.47 from the same dialect region, and 0.19 from the same college province. These figures reveal large heterogeneity in assigned informal tie strength.

Performance. The performance data comprises two main components: (1) monthly call metrics, including the total number of phone calls (call volume) and the number of effective calls (call quality),[8] and (2) subjective evaluations by team leaders assessing each agent's

[8] "Effective" is defined as those calls longer than 1 minute.

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Table 2: Summary Statistics

	Mean	SD	Min	P50	Max
Panel A: Demographics					
Age at hire	31.85	4.92	19	32.00	57
Male	0.93	0.26	0	1.00	1
Education Level	0.56	0.52	0	1	2
Married	0.49	0.50	0	0.00	1
Panel B: First-month performance scores					
Total score	84.03	11.16	27	86.00	100
Call volume score	23.96	8.84	0	25.00	50
Call quality score	6.96	5.00	0	5.00	49
Leader score	18.30	4.24	4	18.00	50
Panel C: Informal ties for a given agent					
# of cohorts in same dorm	2.17	1.53	0	2.00	3
# of cohorts in same dorm and hometown	0.25	0.50	0	0.00	2
# of cohorts in same dorm and dialect region	0.47	0.74	0	0.00	3
# of cohorts in same dorm and college province	0.19	0.49	0	0.00	3

Notes: Education level is coded as 0 for below bachelor's degree (including junior high, senior high, and associate degree, mostly associate degree), 1 for bachelor's degree, and 2 for graduate degree. *Panel C* summarizes informal social ties based on dormitory co-residence: each variable counts the number of randomly assigned cohort-roommates (max 3) who share specific characteristics with the focal agent, such as being from the same hometown province, dialect region, or college province.

effort and professionalism. Although final sales outcomes are also recorded, these occur infrequently. So the company relies heavily on intermediate performance measures for determining bonuses, especially for new hires.

To address privacy concerns and the firm's proprietary interests, we do not use the raw measures (e.g., exact number of calls). Instead, we rely on firm-provided performance scores that are normalized using the company's internal scale. Although the exact normalization method is proprietary, these scores are designed to be comparable across agents. The measures are constructed at the agent–month level and include call volume, call quality, leader evaluation, and a weighted composite of the three (total score). We focus on the first month so that everyone is measured over the same window and to capture short-run productivity effects. Detailed descriptives are shown in Table 2 Panel B.

Tenure and turnover. For each employee, we observe their entry date, exit date (if applicable), and stated reason for leaving. The company rarely fires agents, and almost all

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departures in our data are voluntary quits. The stated reasons for leaving fall into three categories: family obligations (such as caring for elderly parents), external opportunities (such as career changes), and work-related stress. Work-related stress is prevalent, because agents must make numerous daily cold calls to difficult hospital clients with limited organizational or social support. However, we acknowledge that stated reasons may not fully capture true motives for departure.

Tenure is calculated as the number of days between entry and exit. We restrict the sample to workers with an observed exit before the sample cutoff and exclude those still employed at the cutoff (N = 161), whose eventual exit dates are not yet observed. The average realized tenure in this sample is 51 days (SD = 84). To assess whether this restriction introduces systematic bias, we compare tie exposure between censored and non-censored agents across all four tie measures. As reported in Appendix Table E1, none of the differences are statistically significant, suggesting that censoring is unlikely to systematically bias our estimates in either direction.

Motivating Evidence

Before turning to the causal design, we present descriptive patterns that motivate two objects from the conceptual framework: the net retention effect of informal ties and the peer-departure margin.

Fact 1. Dorm residents stay longer. Table 3 compares tenure between agents who choose company housing and those who do not. Column 1 shows that dorm residents stay about 13 days longer on average, roughly 25% of mean tenure. This difference remains similar after adding demographic controls in Column 2. Columns 3 and 4 show a similar pattern when log tenure is used as the outcome, which is the specification we adopt in the later analysis. Because dorm residence is self-selected, we interpret this evidence as descriptive. Even so, the pattern is consistent with the idea that dorm living—where agents are more frequently exposed to informal peer interactions—may strengthen retention.

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Fact 2. Co-resident peers exit around similar times. Agents who share a dorm room with peers from the same cohort tend to leave the firm within similar time windows. Appendix Section Testing for Exit Clustering describes the empirical test in detail. Although descriptive, this clustering pattern suggests that a peer's departure may affect the quitting decisions of remaining agents, consistent with a peer-departure effect. We examine this mechanism formally in Section Peer-Departure Effect.

Table 3: Dorm Residents Stay Longer

	(1) Tenure	(2) Tenure	(3) log(Tenure)	(4) log(Tenure)
Dorm Residence	13.086**	14.407***	0.444***	0.511***
	(5.821)	(4.925)	(0.112)	(0.105)
Male		-18.212*		-0.438***
		(9.986)		(0.139)
Education Level		6.506**		0.160**
		(2.519)		(0.051)
Married		8.854		0.132
		(5.595)		(0.086)
Age at Hire		0.662*		0.015**
		(0.372)		(0.007)
Hometown FE		✓		✓
Cohort FE	✓	✓	✓	✓
N	1,688	1,688	1,688	1,688
R^2	0.071	0.101	0.061	0.108

Notes: Our sample only includes agents who exit during the sample period. Standard errors are in parentheses. Standard errors are clustered at the cohort level. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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EMPIRICAL STRATEGY

This section maps the random dorm assignment design into the three empirical tests implied by our conceptual framework. We first validate the key identifying variation generated by random dorm assignment. We then use assignment-induced exposure to informal ties to conduct the empirical tests identified by our framework. First, we estimate the ITT effect of initial tie exposure on retention, which captures the net effect of informal ties on tenure. Second, restricting attention to the initial onboarding period—when peer departures are

limited—we test whether informal ties reduce early quitting. Third, we study whether the departure of a strong-tie roommate increases the remaining agent's short-run quit hazard.

Random Dorm Assignment

Our identification strategy relies on the assumption that dormitory assignments are randomized conditional on gender. Under this assumption, initial dorm composition, and thus the strength of assignment-induced informal ties, is orthogonal to agents' pre-determined characteristics, alleviating concerns that retention outcomes are driven by sorting into particular rooms. This exogenous variation allows us to separately examine the framework's three key empirical objects: the net retention effect, the present-tie margin, and the peer-departure margin.

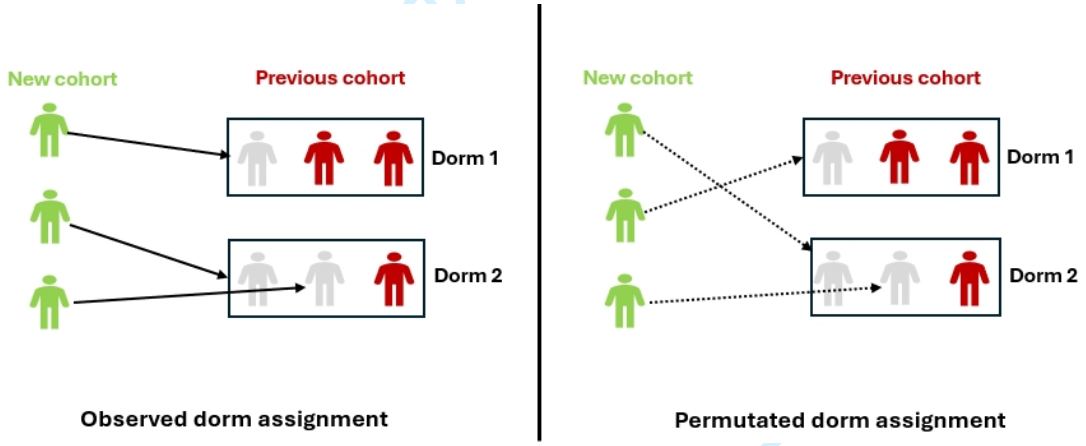

Figure 5: Dormitory Assignment

Note: This figure illustrates the random assignment process for new cohort members to dormitories. The left panel shows the actual assignment where new hires are placed into dorm rooms with varying compositions of previous cohort members. The right panel depicts a hypothetical permutation test, where the same new hires are randomly reassigned across rooms, to generate a benchmark distribution for testing assignment randomness.

We assess the plausibility of conditional random assignment using permutation-based randomization checks. Although the firm describes dorm assignment as random, we still examine whether the realized assignments are consistent with random allocation in the data. For each cohort-dorm combination with at least two agents, we calculate the mean strength of informal ties. For example, in Figure 5, two cohort members are placed in Dorm 2. We

record whether they share the same province (or any other tie measure). If a room houses more than two cohort members, we average the overlap across all pairs.[9]

Next, we simulate 1,000 counterfactual dorm assignments under pure randomness, maintaining the actual dorm room cohort size. Again using the right panel of Figure 5 as an example, we randomly assign two individuals from the same cohort to Dorm 2 and calculate the permuted overlap probability.

We compute overlap rates for five characteristics: three informal tie measures (hometown province, dialect region, and college province) and two demographic variables (age group and marital status). Figure 6 compares the mean observed overlap with the permutation distribution. An overlap rate is the average probability that a roommate shares a given characteristic. For example, an observed hometown overlap of 0.15 indicates that, on average, 15 percent of each agent's same-cohort roommates come from the same province. In all five dimensions, the observed rates lie near the center of the simulated distributions. In other words, the observed rates were statistically indistinguishable from what pure randomness would produce. We thus have more confidence about the random dorm assignment.

Finally, we consider two potential compliance concerns. The first is physical compliance: individuals might negotiate room changes to live with friends, which would weaken the link between initial random assignment and the realized peer environment. However, dormitory records and manager interviews indicate that such negotiations are rare because new hires typically do not know one another before moving into residence.[10] The second is behavioral compliance: even if agents physically reside in their assigned rooms, the assignment matters for retention only if co-residence actually generates meaningful informal ties. We cannot observe tie formation directly in the data, but our survey evidence (presented in Section

[9]Suppose three agents from the same cohort come from Beijing (Person 1), Shanghai (Person 2), and Beijing (Person 3). Person 1 has two same-cohort roommates and her hometown overlap probability is 0.5; person 2's is 0; person 3's is 0.5. The average hometown overlap is $(0.5 + 0 + 0.5)/3 = 0.33$. We then average this statistic across all cohort–dorm cells to obtain the overall overlap probability.

[10]In the most recent cohort, we verified compliance by matching initial assignments to actual residence for 54 dorm residents and found no deviations. Dormitory assignment therefore exhibits near-perfect compliance, and we treat actual room assignment as equal to the initial random assignment.

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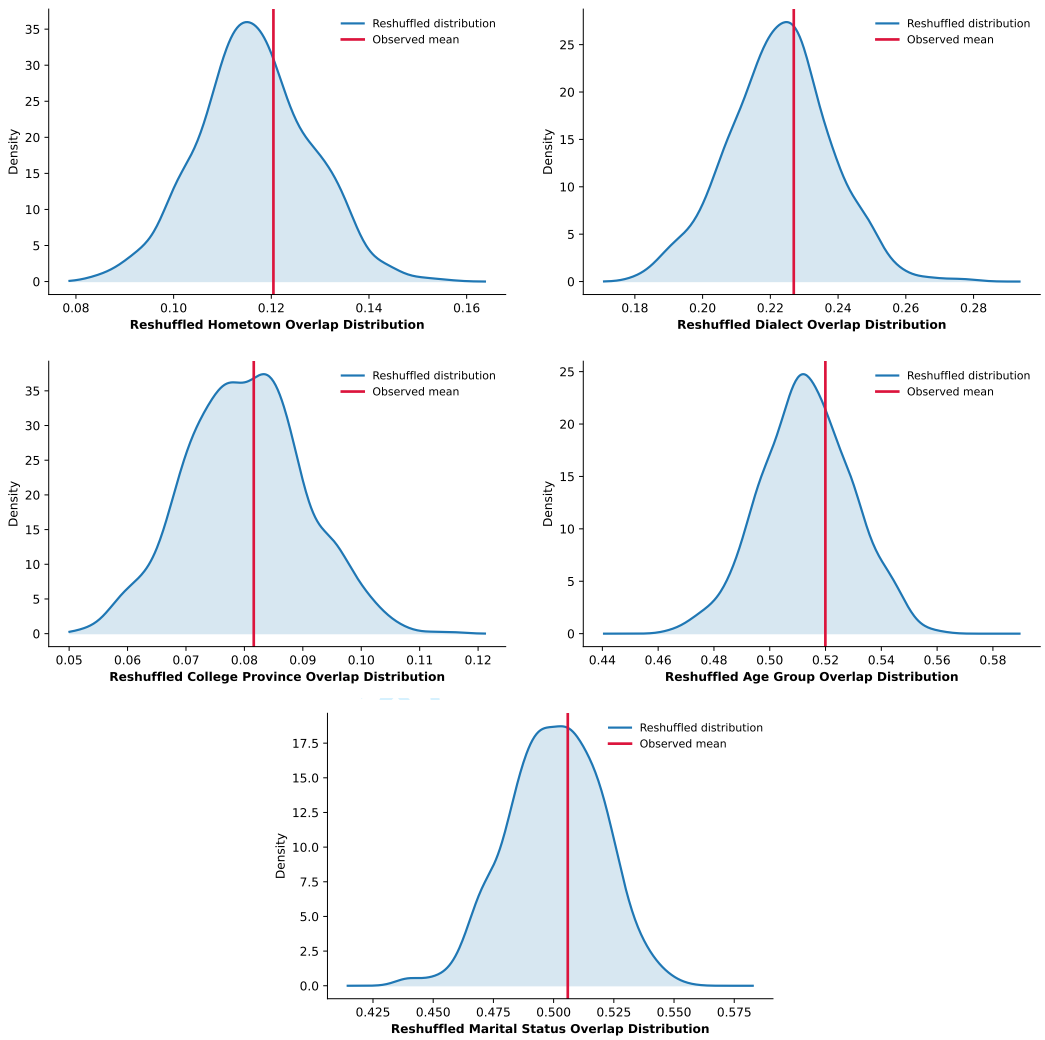

Figure 6: Permutation Test Results for Dormitory Assignment Randomness

Note: The vertical red lines indicate observed overlap rates, which fall well within the distribution of simulated random assignments. Age is grouped into four categories with the following codes: ages 20 or below are coded as 0, ages 21 to 29 as 1, ages 30 to 39 as 2, and ages 40 or above as 3.

Mechanisms) shows that nearly two-thirds of respondents name a dorm roommate as their closest friend in the company (Table 8). Taken together, these patterns provide reassurance that random room assignment translates into genuine informal relationships, rather than merely shared physical space.

Baseline ITT Effects on Tenure

We now provide the estimate of whether assignment-induced informal ties affect retention in total. Specifically, we quantify the ITT effect of being assigned more (and more similar) same-cohort roommates on an agent's tenure. As discussed in the conceptual framework, the reduced-form effect of initial tie exposure $Z_{i(c)}$ on total tenure reflects both the present-tie margin and the peer-departure margin accumulated over the agent's employment spell (Test 1).

We restrict the main estimation sample to agents who opted into company housing. We also focus on agents whose exit is observed before the sample cutoff. This restriction simplifies the analysis by focusing on completed employment spells and avoiding right-censoring. Consistent with our framework, we run the following regression model:

$$\log(\text{Tenure})_{i(c)} = \alpha \cdot Z_{i(c)} + \delta_c + \epsilon_i \tag{7}$$

where $\log(\text{Tenure})_{i(c)}$ is log of days between hire and exit date for agent i in cohort c. $Z_{i(c)}$ captures the agent's exposure to informal ties generated by dorm assignment, and is primarily measured by the number of same-cohort roommates. To capture finer dimensions of cultural proximity, we also construct narrower measures based on shared background characteristics: the number of same-cohort roommates from the same hometown province, the number of same-cohort roommates from the same dialect region, and the number of same-cohort roommates who attended college in the same province. These counts vary solely because of the random dorm assignment and therefore represent exogenous shifters of informal tie intensity. δ_c is the cohort fixed effect, so identification comes from within-cohort variation in roommate composition. ϵ_i is the error term and we cluster standard errors at the cohort level to account for correlation within cohorts.

Identification. Our treatment variable $Z_{i(c)}$ captures the *assigned* (or intended) strength of an agent's informal tie at the moment of dorm assignment. Its coefficient therefore has

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an ITT interpretation. It tells us, for example, how much an exogenous addition of one hometown roommate shifts an agent's expected tenure, regardless of how long that roommate actually stays. Because realized exposure to informal ties may differ from assigned exposure, this ITT need not equal the effect of *realized* informal tie strength. Nevertheless, the ITT is policy-relevant because managers can directly set the initial dorm mix.

Results. Table 4 reports the total effect of each type of informal tie. Column (1) shows that, holding cohort fixed, each additional same-cohort roommate raises expected tenure by 16.1 percent (exp(0.150)-1). Columns (2)–(4) show that an extra same-cohort roommate from the same hometown province raises tenure by 24.6 percent; sharing the same dialect adds 13.2 percent; having attended college in the same province contributes 24.9 percent. All coefficients are statistically significant at least at the 5% level. Across all specifications, the estimated effects are sizable, indicating that informal ties generate economically meaningful retention benefits. Informal ties therefore represent an important determinant of salesperson turnover.

We next assess the robustness of our findings along two dimensions. First, we address a concern that rooms with more assigned same-cohort roommates may have vacancies because of a feature of the room—that is, they are "bad" rooms with lower dorm quality or weaker social climate. If so, stronger informal ties, measured by the number of same-cohort roommates, would be mechanically correlated with worse underlying room conditions and thus with shorter tenure. This concern would bias our estimated effects downward (i.e., lower tenure). To address this concern, Appendix Table E2 controls for the number of incumbents in each room, which proxies for prior occupancy. The results remain virtually unchanged, indicating this concern is not sizable.

Second, to isolate the *incremental* contribution of cultural similarity (hometown, dialect, college) beyond the general retention benefit of entering the firm in the same hiring cohort, Appendix Table E3 reports an alternative specification. In all columns, we include the

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Table 4: Main Effects of Informal Ties on Retention

	(1)	(2)	(3)	(4)
		log(Tenure)		
# roommates:				
From same cohort	0.150***			
	(0.018)			
From same cohort & hometown province		0.223**		
		(0.102)		
From same cohort & dialect region			0.124**	
			(0.059)	
From same cohort & college province				0.222**
				(0.084)
Cohort FE	✓	✓	✓	✓
N	1,083	1,083	1,083	1,083
R^2	0.079	0.059	0.056	0.058

Notes: This table presents estimates examining the causality between informal ties and retention duration (in days) at the company. The sample consists of agents who opted to live in dormitories and exited the firm during the sample period. Standard errors (clustered at the cohort level) are in parentheses. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

number of same-cohort roommates, which captures the baseline retention benefit of being surrounded by same-cohort peers. We then add indicators for whether those same-cohort roommates also share a hometown province, dialect region, or college province with the focal agent. The incremental coefficients are imprecisely estimated because only a subset of same-cohort roommates share these deeper ties, which reduces effective variation and introduces multicollinearity with the baseline same-cohort measure. The baseline same-cohort variable captures most of the variation in tenure (its coefficient remains stable across specifications), and the incremental coefficients are imprecisely estimated due to limited within-cohort variation in deeper ties. We note that the point estimates for shared hometown and college province are directionally positive but cannot draw firm conclusions given the standard errors.

We also examine heterogeneity in Appendix Section Effect Heterogeneity to explore whether the retention effects of informal ties vary across employee characteristics. Guided by the idea that informal social support may be especially valuable for workers facing higher

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adjustment costs or weaker outside options, we estimate subgroup effects by gender, marital status, age, and education. Although the point estimates are generally larger for women, single agents, younger hires, and less-educated workers, the between-group differences are not statistically significant. We therefore do not draw positive conclusions from these patterns.

Overall, the ITT evidence indicates that stronger assignment-induced informal ties increase salesperson retention. However, as discussed, the baseline ITT aggregates both the present-tie margin (ρ) and the peer-departure margin (λ). The sign of the ITT effect is not, by itself, informative about whether informal ties primarily function as support or as competition. We therefore turn to the empirical tests for the present-tie and peer-departure margins separately in Sections Present-Tie Effect (test 2) and Peer-Departure Effect (test 3).

Present-Tie Effect

Our framework implies that early quit behavior can identify the sign of ρ, the present-tie effect parameter, when peer exits are rare. We therefore focus on the initial weeks after dormitory assignment—a period in which both peer exits and self-exits are rare, so that most agents remain on a relatively flat portion of the survival curve. In this early window, any effect of assigned informal ties on quitting should primarily reflect the present-tie effect rather than the disruption caused by peer departures.

We estimate a discrete-time hazard model using a linear probability specification. Specifically, we construct a person-week panel covering the first three ($W = 3$) weeks after hire. Let Quit_{it} equal one if agent i exits in calendar week t, and zero otherwise. We estimate:

$$\text{Quit}_{it} = \beta \cdot Z_{i(c)} + \delta_{r(i)} + \delta_{c(i)} + \delta_{\tau(i,t)} + \varepsilon_{it} \tag{8}$$

where $Z_{i(c)}$ is the assignment-time measure used in Equation 7, capturing the assigned strength of agent i's informal tie at entry. The room fixed effects $\delta_{r(i)}$ absorb time-invariant room-level heterogeneity; cohort fixed effects $\delta_{c(i)}$ absorb cohort-level onboarding conditions;

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and tenure-week fixed effects $\delta_{\tau(i,t)}$ flexibly capture the baseline quit hazard at each tenure stage within the early window. Because departures are rare during this period, $Z_{i(c)}$, the total tie strength at entry, should serve as a close proxy for the theoretical concept R_{it}, the stock of present ties.

Identification. Identification comes from random dorm assignment within cohort, which generates exogenous variation in $Z_{i(c)}$ at entry. The outcome Quit_{it} varies across agents within a given tenure week: among agents who remain employed at the start of tenure week τ, some quit and others do not. Tenure-week fixed effects absorb the average hazard at each tenure stage, so β is identified by comparing agents at the same tenure stage (same τ) who were assigned different levels of informal ties. Under random assignment, these within-tenure comparisons recover the ITT effect of stronger initial ties on early quitting risk.

Results. Table 5 reports estimates of the present-tie margin during the onboarding period. Consistent with informal ties playing a supportive role when present, stronger assigned informal ties at dorm assignment predict significantly lower early quitting. Quantitatively, one additional same-cohort roommate reduces the weekly quit probability by 1.2 percentage points. The effects are stronger when ties are based on more socially salient similarities: one additional hometown-matched same-cohort roommate reduces weekly quitting by 3.6 percentage points, one additional dialect-matched same-cohort roommate reduces weekly quitting by 2.1 percentage points, and one additional college province-matched same-cohort roommate reduces weekly quitting by 1.8 percentage points. All coefficients are statistically significant at the 5% level except the last one, which is marginally significant at 10%. Overall, the pattern supports $\rho > 0$: informal ties provide support during early adjustment and thereby lower short-run quit risk.

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Table 5: Early-Tenure Effects of Informal Ties on Weekly Quit Hazards

	Dependent variable: Quit dummy			
	(1)	(2)	(3)	(4)
# roommates: From same cohort	-0.012**			
	(0.005)			
# roommates: From same cohort & hometown province		-0.036***		
		(0.012)		
# roommates: From same cohort & dialect region			-0.021***	
			(0.007)	
# roommates: From same cohort & college province				-0.018*
				(0.010)
Cohort FE	✓	✓	✓	✓
Room FE	✓	✓	✓	✓
Tenure-week FE	✓	✓	✓	✓
Observations	2 968	2 968	2 968	2 968
R^2	0.047	0.047	0.047	0.045

Notes: This table reports early-tenure (first 3 weeks after hire, $W = 3$) person-week regressions of weekly quitting on informal ties measured at dorm assignment. Standard errors in parentheses are clustered at the cohort level. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Peer-Departure Effect

Test 3 from our framework concerns the peer-departure margin: when a tied peer exits, does quit risk rise (consistent with the support mechanism, $\lambda > 0$) or fall (consistent with competition, $\lambda < 0$)? We therefore examine whether the disruption of an informal tie through a roommate's departure increases an agent's subsequent likelihood of quitting. Consistent with this peer-departure margin, Section Motivating Evidence documents that departures are clustered within dorm rooms. Here, we quantify the magnitude of the effect using a parametric framework and estimate the following discrete-time hazard model:

$$\text{Quit}_{it} = \lambda \, \text{Peer Quit}_{it}^{2w} + \rho \, \text{R}_{it} + \alpha_i + \gamma_{m(t)} + \varepsilon_{it}, \tag{9}$$

where Quit_{it} equals 1 if agent i quits in calendar week t, and $\text{Peer Quit}_{it}^{2w}$ equals 1 if a strong informal-tie peer quits in week $t-1$ or $t-2$. In the baseline specification, peers are defined as same-cohort roommates, who form informal ties through random dorm assignment within cohort. We then estimate separate specifications in which the departing peer also

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shares a particular similarity dimension with agent i, namely hometown province, dialect, or college province. The agent fixed effects α_i absorb time-invariant heterogeneity, such as baseline quit propensity, while the calendar-month fixed effects $\gamma_{m(t)}$ absorb common shocks, including seasonality, policy changes, and macroeconomic conditions.

R_{it} captures the contemporaneous stock of same-type strong-tie peers who remain employed and reside in the dormitory in week t. Including R_{it} allows us to separate two distinct margins. First, a roommate's exit mechanically reduces the stock of remaining informal ties available to the focal agent. Second, the exit may generate an additional short-run *departure shock* that goes beyond this mechanical loss. In this specification, R_{it} absorbs the first margin, while λ captures the second: the incremental effect of recently losing a tied peer, above and beyond the decrease in the number of remaining ties. Because the specification includes agent fixed effects, within-agent variation in R_{it} reflects deviations from an agent-specific average rather than the overall level of present ties emphasized in the conceptual framework. We therefore treat R_{it} as a control variable rather than a focal measure in this analysis.

Identification. Because our focus is on departures, this design does not rely on roommate departures to be exogenous. Instead, identification leverages variation in the timing of departures. We assume that, after controlling for time-invariant agent heterogeneity, calendar-month shocks, and the contemporaneous stock of remaining strong ties, the exact week in which a roommate exits is not systematically related to unobserved shocks to the focal agent's quit risk over the subsequent two weeks. Under this condition, λ identifies the incremental short-run departure shock, rather than the mechanical reduction in present informal ties caused by the exit itself.

Results. Table 6 reports estimates of the short-run departure effect. Column (1) shows that when a same-cohort roommate leaves, the focal agent's probability of quitting within the subsequent two weeks increases by 8.1 percentage points. Columns (2)–(4) examine stronger ties along additional similarity dimensions. Losing a same-cohort roommate from the same

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hometown province increases the weekly quit probability by 12.7 percentage points; the corresponding effect is 10.1 percentage points for a same-dialect roommate and 4.5 percentage points for a roommate who attended college in the same province. Although magnitudes vary across tie types, the estimates are economically meaningful and statistically significant. Overall, the results indicate that tied-peer departures generate a short-run increase in quit risk, consistent with a positive departure parameter ($\lambda > 0$) under the support mechanism. Substantively, this pattern suggests that a peer's exit creates a salient disruption that goes beyond the loss of one present tie, for example by disrupting emotional or informational support routines and making quitting more salient to the remaining agent.

Appendix Table E4 examines whether social similarity amplifies the departure effect beyond the baseline same-cohort effect, using a specification parallel to Appendix Table E3. Though we see a similar weakening in statistical significance, hometown similarity continues to be significant, whereas dialect and college-province overlap yield insignificant effects in this incremental-effects specification.

Summary. Overall, the results suggest that informal ties operate primarily through support rather than competition. First, stronger exposure to informal ties reduces early quit risk, consistent with a positive present-tie effect ($\rho > 0$). Second, the departure of a tied peer increases an agent's subsequent likelihood of quitting, consistent with a positive peer-departure effect ($\lambda > 0$): the exit of a supportive peer disrupts previously available support beyond the mechanical reduction in remaining ties. Together, these findings imply that informal ties improve retention while tied peers remain present, but that their disruption creates a short-run vulnerability to exit. The positive ITT effect of initial tie exposure on overall tenure further suggests that, on average, the cumulative benefits of informal ties when tied peers remain present outweigh the losses associated with later peer departures. Table 7 summarizes how these empirical findings map to the tests in the conceptual framework.

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Table 6: Short–run Effects of Roommate Exits on Weekly Quit Hazards (Controlling for Remaining Peer Stock)

	Dependent variable: Quit dummy			
	(1)	(2)	(3)	(4)
Same-cohort roommate quit	0.081***			
	(0.013)			
Same-cohort roommate quit (same hometown)		0.127***		
		(0.032)		
Same-cohort roommate quit (same dialect)			0.101**	
			(0.031)	
Same-cohort roommate quit (same college province)				0.045*
				(0.020)
Stock of remaining peers	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓
Month FE	✓	✓	✓	✓
Observations	9,286	9,286	9,286	9,286
R^2	0.195	0.191	0.192	0.189

Notes: This table reports discrete-time hazard model estimates of how a roommate's exit affects the focal agent's quit probability within a two-week window. Each specification includes a control for the contemporaneous *stock* of remaining peers of the same type. Standard errors in parentheses are clustered at the month level. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Mapping Empirical Findings to the Conceptual Framework

Conceptual object	Empirical test	Finding	Interpretation
Present-tie effect (ρ)	Early quit hazard	Initial tie exposure lowers early quit risk	$\rho > 0$: informal ties are supportive while present
Peer-departure effect (λ)	Post-departure quit hazard	Peer departure increases subsequent quit risk	$\lambda > 0$: tie disruption raises exit risk beyond lower number of ties
Net retention effect	Tenure regression	Initial tie exposure increases tenure	The benefits of ties while present outweigh the losses from later peer departures

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DISCUSSION

In this section, we first examine the mechanisms through which informal ties enhance retention. We then test whether these benefits extend beyond dormitory networks to formal sales teams, clarifying the boundary of the effect. Finally, we translate our estimates into a back-of-the-envelope cost–benefit analysis to illustrate the managerial implications.

Mechanisms

Why do dorm-based informal ties reduce turnover? Taken together, our empirical tests are consistent with a support-based mechanism: Assignment-induced informal ties improve retention on net, intact ties reduce early quitting, and the departure of a tied peer raises short-run quit risk. To provide more direct evidence on this interpretation, we draw on qualitative survey responses that highlight emotional support as a key benefit of dorm-based relationships. We then ask whether this support operates indirectly through improved performance, and find little evidence for that channel.

Psychological support channel. Informal ties may reduce turnover by directly providing emotional or psychological benefits. To provide further evidence on this channel, we fielded a short survey and conducted semi-structured interviews with current sales agents who opted for dorm living. We examine how roommate relationships shape well-being, belonging, stress management, and intentions to quit. We received 30 completed questionnaires and held in-depth interviews with 8 agents spanning varied tenure, performance, and dorm assignments. Appendix Section Survey and Interview Materials describes the survey design and interview protocol in detail. Below, we summarize the key takeaways.

First, we validate that roommates are the primary friendship network (see Table 8). Nearly two-thirds of respondents (63%) named a dorm roommate as their "closest friend" in the company, far above training-cohort colleagues (23%) and formal teammates (13%). Leisure time is likewise spent mainly with roommates (43%) rather than with teammates

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Table 8: Summary of Survey Responses (N=30)

Survey Item	Count (n=30)	Percentage (%)
Closest friends at the company		
Dormitory roommates	19	63.3
Training cohort colleagues (not roommates)	7	23.3
Team colleagues	4	13.3
Feel most comfortable or familiar with		
Dormitory roommates	20	66.7
Training cohort colleagues (not roommates)	6	20.0
Team colleagues	4	13.3
Biggest challenges/sources of dissatisfaction		
Pressure from high workload or client interactions	26	86.7
Being far from home	16	53.3
Challenging performance appraisal	14	46.7
Turn to for support when facing work-related stress		
Dormitory roommates	20	66.7
Team colleagues	4	13.3
Supervisor/manager	2	6.7
Friends/family outside the company	4	13.3
Prefer to spend leisure time on weekends with		
Dormitory roommates	13	43.3
Team colleagues	4	13.3
Both equally	2	6.7
Neither (prefer time alone or with non-company friends/family)	11	36.7

(13%). These figures confirm that the dorm lottery does, in practice, generate strong informal friendships. This also provides evidence of behavioral compliance: the initial random assignment translates into the meaningful informal ties on which our identification strategy relies.

Second, we find that dorm living creates a shared identity and sense of belonging. When asked whom they felt "most familiar and comfortable with", 67% again chose roommates. Interviewees explained that arriving in an unfamiliar city makes co-resident peers feel "like family". One newcomer (9 months tenure) remarked: "We were all new and facing the same challenges. My roommates understood me in ways my teammates or leader couldn't." Such

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accounts illustrate how dorm life supplies the social anchoring that is valuable during the high-risk early months.

Third, we find roommates buffer work stress. The job is pressure-filled: 87% of respondents cited "high workload or difficult client interactions" as their biggest source of stress. Asked whom they turn to when stressed, 67% selected roommates, compared with just 13% for teammates and 7% for supervisors. As one agent (5 months tenure) put it: "After a bad sales day, I don't want to face my team leader. With my roommates, especially those who speak the same dialect, I can vent without judgment. Without them I'd have quit two months ago."

Overall, the survey evidence supports the psychological support mechanism. Interviewees consistently emphasize the value of emotional support and a sense of belonging, which help translate dorm-based informal ties into retention benefits.

Performance channel. Psychological support could also reduce turnover indirectly—for example, if informal ties help newcomers adjust and perform better early on. Because performance positively predicts retention (see Appendix Figure E3), such an effect would translate into longer tenure. We therefore ask whether informal peer ties improve retention by raising performance (productivity).

Leveraging the dorm lottery, we re-estimate our main specification (Equation 7) with performance as the dependent variable (Table 9). Across all informal-tie measures, the estimated coefficients are small and statistically insignificant, providing no evidence consistent with an indirect productivity-gain channel. One caveat is that performance data are available only for agents who remain beyond the first month, so this analysis necessarily conditions on survival. If early leavers are disproportionately weakly connected, as suggested by our main results, and would also have performed poorly (see Appendix Figure E3), their exclusion would attenuate the estimated relationship between informal ties and performance. We therefore interpret the null performance estimates as suggestive. Importantly, this caveat

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applies only to the mechanism test and does not affect our main retention estimates.

Taken together, these results suggest that informal ties reduce turnover primarily through direct psychological support—such as buffering stress and strengthening social attachment—with limited evidence that indirect productivity gains are the main mechanism.

Table 9: Effects of Informal Ties on Performance

	(1) Total Score	(2) Total Score	(3) Total Score	(4) Total Score
# roommates:				
From same cohort	-0.001			
	(0.003)			
From same cohort & hometown province		-0.005		
		(0.013)		
From same cohort & dialect region			-0.007	
			(0.008)	
From same cohort & college province				-0.011
				(0.012)
Cohort FE	✓	✓	✓	✓
N	540	540	540	540
R^2	0.042	0.032	0.036	0.031

Notes: This table presents estimates examining the causal effects of informal ties with dorm roommates on first-period performance at the company. We have fewer observations here because performance data are available only for agents who remain beyond the first month; thus, the analysis conditions on survival. Standard errors are clustered at the cohort level. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Informal Ties within Formal Teams

We next examine whether the retention effects of informal ties extend beyond dormitories to formal sales teams. Within our conceptual framework, the team setting provides a useful boundary test: the effect of informal ties depends on the balance between support and competition, and that balance may differ across organizational contexts. Relative to dormitories, formal teams involve greater performance pressure, closer monitoring, and more direct competition for recognition. These features may weaken the support channel and strengthen the competition channel. If so, the net retention effect of informal ties in teams should

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be smaller—or even negative—relative to the positive effect we document in the dormitory setting.

To test this implication, we therefore replace dorm-level measures with team-level analogues: the number of teammates who share an agent's hometown province, dialect region, or college province, plus indicators for having a team leader from the same hometown or dialect region.[11] The firm reports that since 2022, new hires have been assigned to teams by lottery to reduce favoritism. Appendix Section Random Assignment to Teams details our identification checks following Guryan et al. (2009) and provides evidence that team assignments are unrelated to employees' pre-existing social ties.

Table 10: Effects of Ties within Formal Teams on Retention

	(1)	(2)	(3)	(4)	(5)	(6)
			log(Tenure)			
# teammates:						
From same cohort	0.003					
	(0.021)					
From same cohort & hometown province		-0.036				
		(0.052)				
From same cohort & dialect region			-0.059**			
			(0.028)			
From same cohort & college province				0.012		
				(0.047)		
With the team leader:						
From same hometown					-0.033	
					(0.074)	
From same dialect region						-0.055
						(0.072)
Cohort FE	✓	✓	✓	✓	✓	✓
N	1,688	1,688	1,688	1,688	1,688	1,688
R^2	0.035	0.035	0.038	0.035	0.035	0.035

Notes: This table presents estimates examining the causal effects of ties within formal teams on retention duration (in days) at the company. Standard errors are in parentheses. Standard errors are clustered at the cohort level. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10 shows that across all specifications, coefficients are small, mostly negative, and

[11] Team-leader education is unavailable.

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statistically insignificant. The exception is sharing a dialect region with teammates, which *shortens* tenure by about 5.9 percent (column 3); this effect runs opposite to the pattern observed in dorms.

These null results suggest that the positive retention effect of informal ties we document in dormitories does not extend to formal sales teams in our setting. Within our framework, this pattern is consistent with formal team environments shifting the balance from support toward competition. Teammates may be viewed primarily as professional collaborators rather than close personal contacts, limiting the mutual reassurance and empathy that dorm-based ties can generate. Although this interpretation may be context-specific, the contrast between dormitories and teams reinforces our broader conclusion that the retention benefits of roommate ties operate primarily through a psychological support channel. Future research could examine the organizational contexts in which formal team ties generate similar forms of support and, in turn, improve retention.

Managerial Implications

Back-of-the-envelope savings from homophilous dorm assignment. How much annual cost savings could the firm realize by systematically assigning new hires who share backgrounds to the same dorm room, instead of relying on the current random lottery system? To give managers intuition, we translate our ITT estimate from Table 4, which indicates that adding one same-cohort, same-hometown roommate boosts tenure by roughly 25 percent, into recruitment and training cost savings.

Table 11: Input Parameters for the Back-of-the-Envelope Calculation

Input	Symbol	Value	Source
New hires who live in dorms each year	N	700	Our raw data
Baseline mean tenure of new hires	T_0	7 weeks	Our raw data
Tenure gain from +1 cohort & hometown roommate	$\Delta\%$	+25%	Table 4
Recruiting + training cost per hire	$C_{\text{hire+train}}$	25,000 RMB	HR cost ledger

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Table 11 summarizes the inputs for our back-of-the-envelope calculation. We begin by translating the estimated intention-to-treat (ITT) effect into an implied increase in tenure per hire. Let $\Delta\%$ denote the proportional increase in tenure induced by stronger informal ties, and let T_0 denote baseline average tenure. Using $\Delta\% = 0.25$ and $T_0 = 7$ weeks, the implied tenure gain is

$$\Delta T \;=\; \Delta\% \times T_0 \;=\; 0.25 \times 7 \text{ weeks} \;=\; 1.75 \text{ weeks}.$$

Next, we calculate the implied reduction in annual hiring needs. Let N denote the number of new hires the firm typically recruits each year. Because an increase of ΔT weeks represents a $\Delta T/T_0$ proportional reduction in turnover, the number of hires avoided is

$$\Delta\text{Replacements} \;=\; N \times \frac{\Delta T}{T_0} \;=\; 700 \times \frac{1.75}{7} \;=\; 175 \text{ hires}.$$

Finally, we compute the resulting cost savings. Let $C_{\text{hire+train}}$ denote the per-hire recruiting and training cost. Multiplying the reduction in hires by this unit cost yields annual savings of

$$\text{Annual savings} \;=\; \Delta\text{Replacements} \times C_{\text{hire+train}} \;=\; 175 \times 25{,}000 \text{ RMB} \;=\; 4.38 \text{ million RMB}.$$

The figure of 4.38 million RMB (approximately 8.7% of the firm's annual profit) is the annual reduction in recruitment and training cost if management guarantees each newcomer one additional same-cohort, same-hometown roommate, relative to today's random assignment. This is a conservative estimate of total cost saving: we exclude productivity losses while a position is vacant and any sales revenue effects. Because dorm reassignment entails negligible administrative cost and requires no changes to compensation, training, or work design, it is arguably among the most cost-effective retention levers available to the firm.

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Broader implications. Our findings have broader implications for firms seeking to reduce sales agent turnover. More fundamentally, the conceptual framework we develop—distinguishing present-tie effects from peer-departure effects—offers a practical diagnostic roadmap that firms can apply in their own settings. The appropriate managerial response depends on two questions: which margin dominates, and whether informal ties operate through support or competition. Diagnosing both is a prerequisite for effective intervention.

First, the social environment facing new hires can matter substantially for retention. In our setting, improving early social matching among newcomers yields retention gains equivalent to approximately 8.7% of annual profit—and because dormitory reassignment entails negligible administrative cost and requires no changes to compensation, training, or work design, it is among the most cost-effective retention levers available. While the precise magnitude will vary across organizational contexts, the direction of the effect is likely to generalize: onboarding arrangements that facilitate early peer connection—whether through training cohorts, temporary co-location, or repeated informal interaction among newcomers—can be expected to yield meaningful retention benefits.

Second, our results indicate that the retention value of informal ties operates primarily through psychological support rather than improved productivity. This suggests that retention policies may be more effective when they address the social experience of new hires alongside traditional investments in skills and incentives—particularly in high-pressure sales environments where adjustment costs are high and peer reassurance is valuable.

Third, peer departures can create an additional short-run retention risk beyond the mechanical loss of remaining ties. This implies that firms may benefit from treating early departures as signals of elevated risk within local informal networks. Light-touch responses—such as check-ins with affected peers, temporary workload relief, or added support resources—may help contain subsequent exits. More broadly, the results suggest that informal networks are not merely a background feature of the workplace, but an economically meaningful part of retention policy.

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Finally, although our specific estimates are context-dependent, any firm can apply the framework to diagnose the role of informal ties in their own environment. If data on peer composition and exit timing are available, firms can estimate both margins and assess whether ties operate through support or competition—then calibrate their response accordingly: investing in tie formation if supportive present ties dominate, prioritizing departure management if the peer-departure effect is large, or reconsidering tie cultivation altogether if competition dominates. The framework thus functions as both an analytical device and a managerial tool for acting on the social foundations of employee retention.

CONCLUSION

In this paper, we study how informal ties affect salesperson turnover. We develop a conceptual framework that distinguishes the effect of informal ties while tied peers remain present from the effect of peer departures—providing a generalizable measurement approach for diagnosing whether informal ties operate through support or competition in a given organizational setting. Applying this framework to a large sales organization with randomized dormitory assignments, we document three main findings. First, stronger assignment-induced informal ties reduce turnover on net, indicating the retention value of dorm-based peer relationships. Second, the pattern of estimates is most consistent with a support-based mechanism: present ties reduce early quitting, whereas the departure of a tied peer raises short-run quit risk. Third, these retention effects do not appear to operate through improved productivity, pointing instead to psychological support as the primary channel. To the best of our knowledge, this is the first paper to causally identify the effect of informal ties on realized turnover behavior.

Our study points toward two directions for future research. First, our findings stem from a single organizational setting; future work could test whether the same patterns emerge across diverse industries and cultural contexts. Second, although our survey evidence suggests that assigned roommates do form meaningful ties and that these ties provide psychological

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support, we do not observe tie formation or psychological states directly. Richer data on actual social interactions between assigned peers would allow researchers to move beyond intent-to-treat estimates and more directly quantify both how co-residence translates into realized informal ties and through what channels those ties shape retention. That said, the framework and measurement approach we develop travel beyond this setting, offering researchers and managers a practical roadmap for diagnosing the role of informal ties in retention.

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Web Appendix:

Informal Ties and Sales Agent Turnover: Evidence from Randomized Dormitory Assignment

April 22, 2026

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Web Appendix A: Testing for Exit Clustering

This section provides the empirical procedure underlying Figure A1, which examines whether agents sharing a dorm room with same-cohort peers tend to leave the firm in close succession.

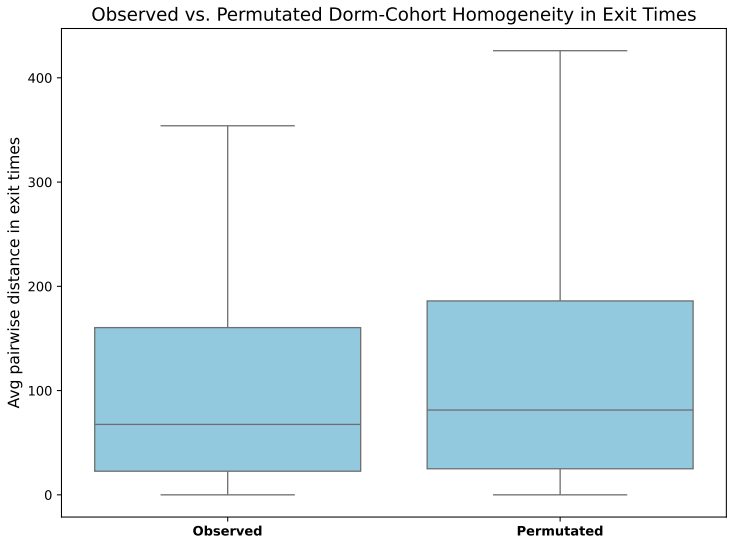

Figure A1: **Exit Time Similarity Within and Across Dormitories**

Note: This figure compares the distribution of average pairwise differences in exit dates within actual dormitory-cohort groupings (535 dorm-cohort groups, left box plot) against the random simulated distribution (right box plot). The lower median and tighter distribution for observed roommate groups demonstrate: co-resident peers from the same cohort tend to exit the company at more similar times than would occur by random chance, suggesting a possible social influence effect on turnover timing.

For every room that contains at least two members of the same cohort, we compute all absolute pairwise differences in exit dates (e.g., A–B, A–C, B–C) and take their average. To construct a benchmark, we run a permutation test: for each observed room-cohort of size n, we randomly sample n agents from the same cohort, ignoring dorm assignments, repeat this 1,000 times, and record the mean pairwise exit gap for each draw.

Figure A1 plots the distribution of these simulated gaps alongside the observed gaps. Exit times for observed dorm-cohort mates are more clustered than the random benchmark, indicating that co-resident peers tend to exit at similar moments. This pattern is consistent with peer-driven contagion.

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Web Appendix B: Effect Heterogeneity

We examine how the retention effect of informal ties varies across agent subgroups. We re-estimate Equation (7) within subsamples defined by key agent attributes: gender, marital status, age at hire, education, distance of hometown from the firm, and distance of college from the firm. Continuous variables are divided at the sample median to create high- and low-groups.

Figure B1 depicts the coefficient on "# roommates from the same cohort" estimated separately for each subgroup, with 95 percent confidence intervals. Point estimates suggest that informal ties confer larger retention benefits for employees who may be more vulnerable in adjustment—female, single, less-educated, younger, or more geographically distant. A natural interpretation is that these groups tend to arrive with fewer pre-existing connections and thus rely more heavily on workplace peers for social support and information, making assignment-induced ties particularly valuable. At the same time, the confidence intervals overlap substantially across groups, and many pairwise differences are not statistically significant. We therefore view the heterogeneity patterns as suggestive; the imprecision likely reflects limited statistical power in subgroup analyses rather than the absence of true underlying heterogeneity.

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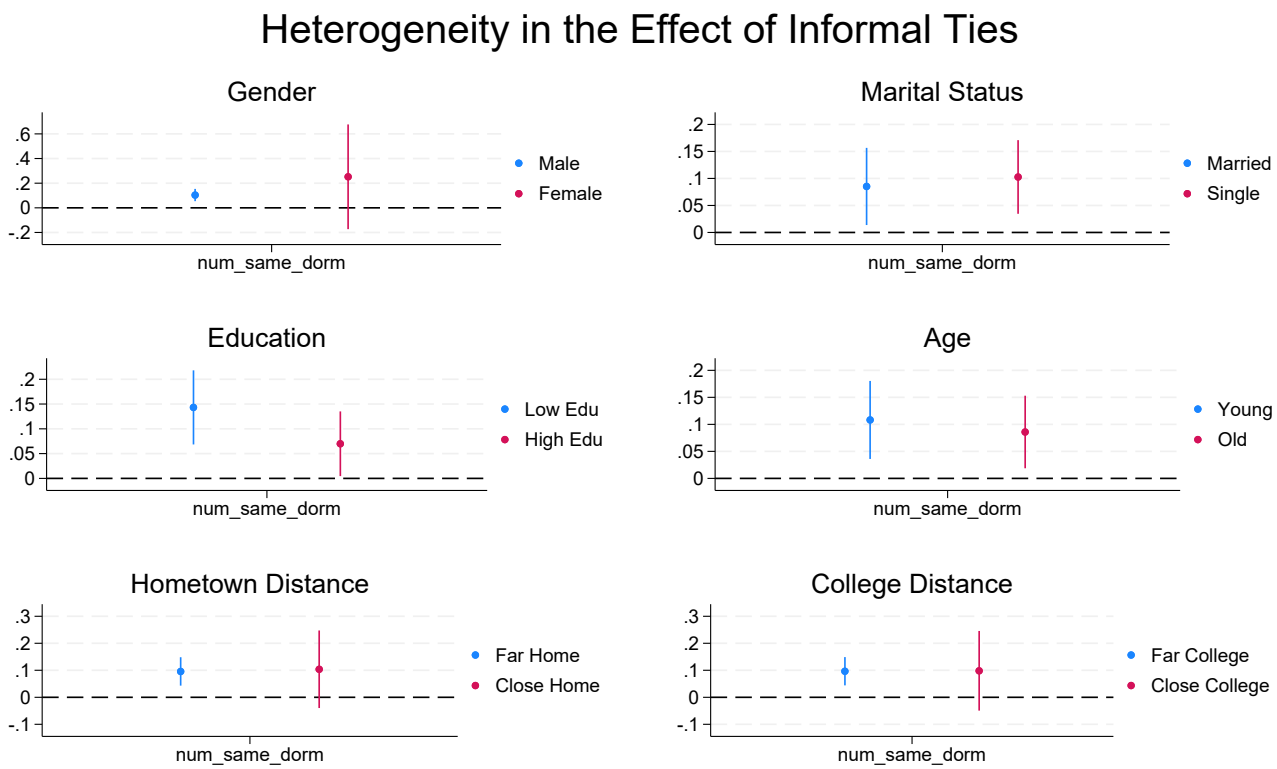

Figure B1: **Effect Heterogeneity**

Note: Each point is the coefficient from a regression of log tenure on # roommates, estimated separately by the listed characteristics. For continuous moderators, we split the sample at the median.

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Web Appendix C: Random Assignment to Teams

Following Guryan et al. (2009), we implement a formal test of random assignment. We regress individual i's pre-determined characteristic x on the average x of i's peers. If selection into peer groups is ignorable, then this regression should yield a coefficient of zero. We take the following form:

$$X_{itg} = \pi \overline{X}_{-i,tg} + \phi \overline{X}_{-i,t} + \delta_t + \epsilon_{itg}$$

Here, X_{itg} is the predetermined characteristic of agent i from training cohort t assigned to team g. $\bar{X}_{-i,tg}$ is the leave-out mean of characteristic X for i's teammates in team g (excluding agent i), and $\bar{X}_{-i,t}$ is the mean of characteristic X across all agents in training cohort t (again excluding agent

i). We include cohort fixed effects δ_t to account for any time-varying factors in the recruitment process. The coefficient of interest is π, which captures any systematic relationship between an individual's characteristics and those of their teammates.

The predetermined characteristics we observe are the agent's gender, age at hire, education level, marital status, and place of birth. These characteristics are determined prior to team assignment and thus should be uncorrelated with team composition under the random assignment. To implement this test, we estimate this equation separately for each characteristic and examine whether the coefficient π is statistically different from zero.

Table C1: **Regression Results: Testing Random Assignment**

	(1) Age	(2) Male	(3) Education	(4) Married	(5) Cantonese	(6) Beijing
Team leave-out-mean (π)	0.009	-0.006	0.045	-0.003	-0.017	0.010
	(0.007)	(0.028)	(0.038)	(0.011)	(0.036)	(0.020)
Cohort leave-out-mean	✓	✓	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓	✓	✓
Observations	1,987	1,987	1,987	1,987	1,987	1,987
R^2	0.880	0.861	0.041	0.850	0.826	0.822
	(7) Hakka	(8) Jianghuai	(9) Northwestern	(10) Southwestern	(11) Northeastern	(12) Zhongyuan
Team leave-out-mean (π)	0.005	0.034	-0.012	0.000	-0.001	-0.013
	(0.020)	(0.018)	(0.023)	(0.025)	(0.022)	(0.017)
Cohort leave-out-mean	✓	✓	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓	✓	✓
Observations	1,987	1,987	1,987	1,987	1,987	1,987
R^2	0.833	0.856	0.822	0.888	0.850	0.862

Notes: This table validates the random assignment of agents to teams by testing whether individual characteristics predict team composition. Each column regresses a different individual characteristic (age, gender, education, marital status, and dialect regions) on the team leave-out-mean of that characteristic, controlling for cohort leave-out-means and cohort fixed effects. The consistently small and statistically insignificant coefficients across all specifications confirm that team assignment is orthogonal to individual characteristics. Standard errors are in parentheses and are clustered by the training cohort.

Table C1 presents the results. The lack of statistical significance on the team leave-out-mean coefficients (π) across all specifications suggests that agents were not systematically assigned based on observable characteristics. The magnitudes are also economically small—for example, a one standard deviation increase in teammates' age is associated with only a 0.009 year increase in an individual's age. Similar patterns hold for other characteristics including gender, education, marital

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status, and various dialect regions, with no coefficient being statistically different from zero at conventional levels. The high R^2 values in most specifications are driven by the cohort fixed effects, reflecting the importance of controlling for time-varying factors in the recruitment process.

Web Appendix D: Survey and Interview Materials

To capture the lived experiences of sales agents, we designed a confidential survey followed by some interviews. Our short survey mainly identifies who, how, why, and when agents form relationships with roommates; and also compares this relationship with formal ties in the workplace. We distributed questionnaires to 32 current sales agents with dormitory experience, receiving 30 completed responses (93.75% response rate). The survey evidence underscores the unique and greater importance of informal ties, consistent with our main findings.

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销售人员宿舍生活与工作关系调查问卷

调查说明： 本次调查旨在了解您在公司宿舍安排和工作场所人际关系方面的体验。所有回答将严格保密，仅用于研究目的。请如实回答所有问题。

---

1. 在公司里，您认为谁是您最亲密的朋友？

- □ 宿舍室友　　□ 同期培训的同事
- □ 团队同事　　□ 其他（请说明）：_____

2. 在公司里，您觉得和谁相处最舒服或最熟悉？

- □ 宿舍室友　　□ 同期培训的同事（非室友）
- □ 团队同事（工作小组）　　□ 其他（请说明）：_____

3. 在公司工作期间，您遇到的最大挑战或不满来源是什么？

- □ 高工作量或客户交流带来的压力　　□ 严格苛刻的绩效考核
- □ 不愉快或困难的同事关系　　□ 离家太远
- □ 缺乏职业发展或晋升机会　　□ 公司政策或流程
- □ 其他（请说明）：_____

4. 当面临工作压力时，您通常会向谁寻求支持？

- □ 宿舍室友　　□ 团队同事　　□ 主管/经理
- □ 公司外的朋友/家人　　□ 其他（请说明）：_____

5. 周末休息时，您更愿意和谁一起度过？

- □ 宿舍室友　　□ 团队同事
- □ 两者都可以　　□ 都不选（更愿意独处或与公司外的朋友/家人一起）

6. 根据您的经验，您如何评价以下人际关系：

a. 宿舍室友之间的关系：□ 非常差　□ 较差　□ 一般　□ 良好　□ 非常好

b. 团队同事之间的关系：□ 非常差　□ 较差　□ 一般　□ 良好　□ 非常好

感谢您的参与！

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Survey Questionnaire Sample (Translated Version)

Survey Instructions: This survey aims to understand your experiences with company dormitory arrangements and workplace interpersonal relationships. All responses will be kept strictly

confidential and used for research purposes only. Please answer all questions honestly.

1. **At the company, who do you consider to be your closest friend?**

   ☐ Dormitory roommates

   ☐ Colleagues from the same training cohort

   ☐ Team colleagues

   ☐ Other (please specify): _____

2. **At the company, who do you feel most comfortable or familiar with?**

   ☐ Dormitory roommates

   ☐ Colleagues from the same training cohort

   ☐ Team colleagues

   ☐ Other (please specify): _____

3. **During your time working at the company, what has been your biggest challenge or source of dissatisfaction?**

   ☐ Pressure from high workload or client interactions

   ☐ Strict and demanding performance appraisals

   ☐ Unpleasant or difficult colleague relationships

   ☐ Being far from home

   ☐ Lack of career development or promotion opportunities

   ☐ Company policies or procedures

   ☐ Other (please specify): _____

4. **When facing work-related stress, who do you usually turn to for support?**

   ☐ Dormitory roommates

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- ☐ Team colleagues
- ☐ Supervisor/Manager
- ☐ Friends/family outside the company
- ☐ Other (please specify): _____

5. **During weekend breaks, who would you prefer to spend time with?**

- ☐ Dormitory roommates
- ☐ Team colleagues
- ☐ Either is fine
- ☐ Neither (prefer time alone or with friends/family outside the company)

6. **Based on your experience, how would you rate the following interpersonal relationships:**

a. Roommates:	☐ Very poor	☐ Poor	☐ Average	☐ Good	☐ Very good
b. Teammates:	☐ Very poor	☐ Poor	☐ Average	☐ Good	☐ Very good

We also conducted semi-structured interviews with 8 agents (10-15 minutes each) to understand dormitory relationships more deeply. The interviews reveal some interesting points. First, roommates create "intervention moments" that prevent quitting. One agent (7 months tenure) recalled: "I had my resignation letter written. My roommate found it and we talked until 3 AM. He reminded me why I came here." Another added: "Three times I packed to leave. Three times my roommates unpacked my bags." Second, shared backgrounds create micro-support networks. As one Sichuan agent explained: "Two of us cook hometown food every Sunday. We joke in our dialect. For those hours, we're home." These cultural connections provided identity anchors in an unfamiliar city. Third, roommates share the same vulnerabilities. A recent hire (3 months tenure) explained: "My team leader says 'push through', but my roommate gets it. We're all pretending to be confident during the day, but at night we can be honest about struggling." This peer-level emotional validation emerges as a critical factor in building resilience during the difficult early months.

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Web Appendix E: Additional Tables and Figures

Table E1: **Tie Exposure: Censored vs. Non-Censored Agents**

	Mean Tie Exposure			
Tie Measure	Censored	Non-Censored	t-statistic	p-value
From same cohort	1.893	2.223	−1.137	0.256
From same cohort & hometown province	0.214	0.250	−0.374	0.708
From same cohort & dialect region	0.321	0.480	−1.121	0.263
From same cohort & college province	0.250	0.192	0.622	0.534

Notes: Censored agents are those still employed at the sample cutoff. t-tests compare mean tie exposure between censored and non-censored agents. None of the differences are statistically significant at conventional levels.

Table E2: **Robustness: Controlling for the Number of Incumbent Roommates**

	(1)	(2)	(3)	(4)
		log(Tenure)		
# roommates:				
From same cohort	0.148***			
	(0.018)			
From same cohort & hometown province		0.221**		
		(0.101)		
From same cohort & dialect region			0.122**	
			(0.057)	
From same cohort & college province				0.222**
				(0.082)
Number of incumbent roommates	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓
N	1,083	1,083	1,083	1,083
R^2	0.080	0.060	0.057	0.060

Notes: Incumbents refer to roommates from earlier cohorts who were already living in the dorm when the focal agent checked in. All regressions control for the number of incumbent roommates and include cohort fixed effects. Standard errors (clustered at the cohort level) are in parentheses. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Table E3: **Incremental Effects of Informal Ties on Retention**

	(1)	(2)	(3)	(4)
		log(Tenure)		
# roommates:				
From same cohort	0.150***	0.139***	0.145***	0.139***
	(0.018)	(0.020)	(0.021)	(0.021)
From same cohort & hometown province		0.123		
		(0.105)		
From same cohort & dialect region			0.027	
			(0.070)	
From same cohort & college province				0.109
				(0.090)
Cohort FE	✓	✓	✓	✓
N	1,083	1,083	1,083	1,083
R^2	0.079	0.081	0.079	0.081

Notes: This table reports regressions that include the baseline same-cohort variable in all specifications to estimate the incremental effect of additional similarities. The dependent variable is log tenure (days). Standard errors clustered at the cohort level are in parentheses. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E4: **Short–run Incremental Effects of Same-Cohort Roommate Exits on Weekly Quit Hazards**

	Dependent variable: Quit dummy			
	(1)	(2)	(3)	(4)
Same-cohort roommate quit	0.080***	0.072***	0.070***	0.083***
	(0.013)	(0.012)	(0.013)	(0.013)
Same-cohort roommate quit (same hometown)		0.064**		
		(0.029)		
Same-cohort roommate quit (same dialect)			0.040	
			(0.031)	
Same-cohort roommate quit (same college province)				−0.023
				(0.020)
Individual FE	✓	✓	✓	✓
Month FE	✓	✓	✓	✓
Observations	9 368	9 368	9 368	9 368
R^2	0.258	0.258	0.258	0.258

Notes: This table reports discrete-time hazard model estimates of the *incremental contagion effect* of same-cohort roommate departures. Standard errors in parentheses are clustered at the month level. Significance codes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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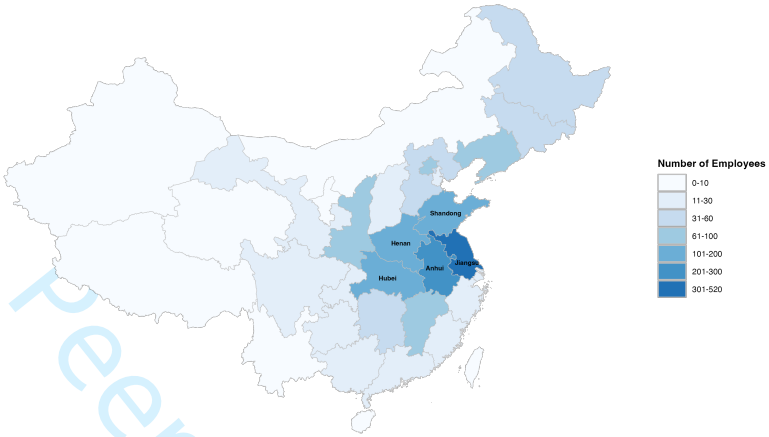

Figure E1: **Employee Distribution by College Province**

Note: This map displays the geographic distribution of sales agents by the provinces where they attended college.

Figure E2: **Employee Distribution by Dialect Region**

Note: This map shows the distribution of sales agents by dialect regions.

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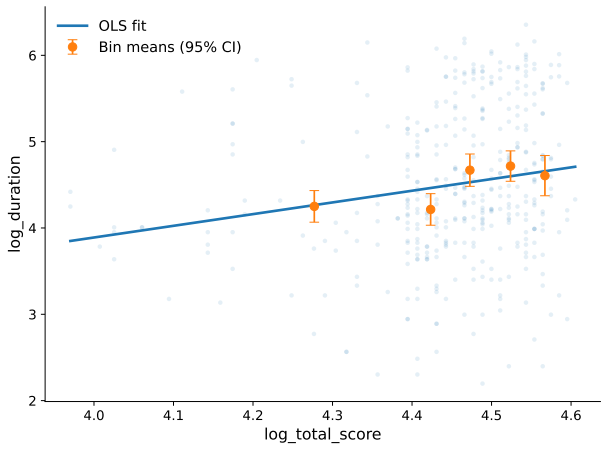

Figure E3: **Relationship Between First-Month Performance and Tenure**

Note: This figure presents a binscatter of agents' subsequent tenure against their first-month performance. Each marker represents the average tenure within a bin of the first-month performance distribution. The solid line is the fitted linear relationship from an OLS regression.

References

Guryan, J., Kroft, K., and Notowidigdo, M. J. (2009). Peer effects in the workplace: Evidence from random groupings in professional golf tournaments. *American Economic Journal: Applied Economics*, 1(4):34–68.

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